

No. of Printed Pages : 4
Roll No.

221562

6th Sem/ Instrumentation & Control
Subject : Biomedical Instrumentation

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 EEG machine is a _____ instrument.
a) Therapeutic b) Clinical laboratory
c) Diagnostic d) None of these
- Q.2 Heart is associated with _____ system.
a) Cardiovascular b) Respiratory
c) Nervous system d) Bio-Chemical
- Q.3 IC-741 _____ pin IC.
a) 8 b) 4
c) 10 d) 14
- Q.4 The material used for bio electrodes are _____.
a) German silver
b) Plastic or Rubber
c) Nylon & stainless steel
d) All of these

- Q.5 Beta waves are associated with _____.
 a) ECG b) EEG
 c) EMG d) None of these
- Q.6 Magnetic waves are used in _____ imaging system.
 a) X-Ray b) CT Scan
 c) MRI d) Ultra sound

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Transducer is a device which converts one form of energy in to another form. (True/False)
- Q.8 Mention one application of X-Ray.
- Q.9 _____ is the approximate range of resting potential.
- Q.10 Expand CMRR.
- Q.11 Glucometer is used to measure blood pressure. True/False
- Q.12 Expand MRI.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Discuss cardiovascular and nervous system.
- Q.14 Draw pin configuration of IC-741.

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- Q.15 Write short note on cell structure.
- Q.16 Explain Action Potential.
- Q.17 Discuss Op-AMP as adder and sub tractor.
- Q.18 Explain ECG Machine.
- Q.19 Write short note on Pacemaker.
- Q.20 Discuss blood pressure measurement.
- Q.21 Describe working principle of X-Ray machine.
- Q.22 Write short note on Pulse Oxy meter.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Draw and explain generalized block diagram of a medical instrumentation system.
- Q.24 Describe OP-AMP as inverting & non-inverting mode and Integrator & differentiator.
- Q.25 Explain working principle of ultra sound machine. Also mention its applications.

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